MLR780 Air Conditioning and Refrigeration 2021

Assignment 2
Thermal Comfort

Instructions

- This individual assignment is to be submitted online before midnight on **25 April 2021**.
- Repetitive calculation will be needed. Use a programming platform or a spread sheet of your choice. Use CoolProp for thermodynamic properties.
- Prepare a technical report that supplies all relevant equations, methods and input values, and results. Include everything that is to be graded in your report. Name it: "MLR780-2021-Assignment1-SURNAME.pdf"
- Each submission (PDF) **must** be accompanied by an official cover page obtainable from Click-UP, **as well as** your supporting documentation to prove you performed the analysis.
- The supporting documentation will not be graded, but will be checked: It must be a soft version of the script (a *.txt text file) or spread sheet with the following naming convention: "MLR780-2021-Assignment1-SURNAME.* " (The "*" represent the file extension, which will depend on the program you used – it should not be "pdf").
- **Only the PDF report will be graded.**
- Keep a time sheet documenting the time it took you to solve the questions and indicate this on the first page of your submission. Time spent on the lecture content / example / programming revision should not be included.

Outcomes:

This assignment has been designed to assist you to understand the fundamentals of mean radiant temperature, operative temperature, how to calculate this from first principles and how to apply these to an arbitrary space to make a judgement on thermal comfort of an occupant.

Background:

Consider Figure 1 showing a simple office space 11 m long (L) in the z direction and 5 m deep (B) in the x direction. The ceiling height is 3 m (H). The space is fully enclosed by walls on three sides and an external glazed[1] wall on the fourth side. A section of the roof (width b = 2 m) is fully glazed. Line 1 is 1 m from the rear long wall and line 2 is 1 m from the front long glazed wall.Table 1 contains measured semi steady-state surface temperatures (on the room side) for a moment in time. PLEASE USE THESE WALL NUMBERS WHEN YOU TABULATE YOUR RESULTS. The office is located at altitude with an atmospheric pressure of 89 kPa (p_t). A split unit air conditioning system maintains the indoor air temperature at 24°C (t_i). The relative humidity is 40% (ϕ_i).

Because the different surfaces of the room are at different temperatures, it will result in different directional mean radiant temperatures for an occupant. The overall mean radiant temperature and operative temperature will also be different depending on the location in the space.

[1] Glazed = surface which is transparent / translucent (such as glass).

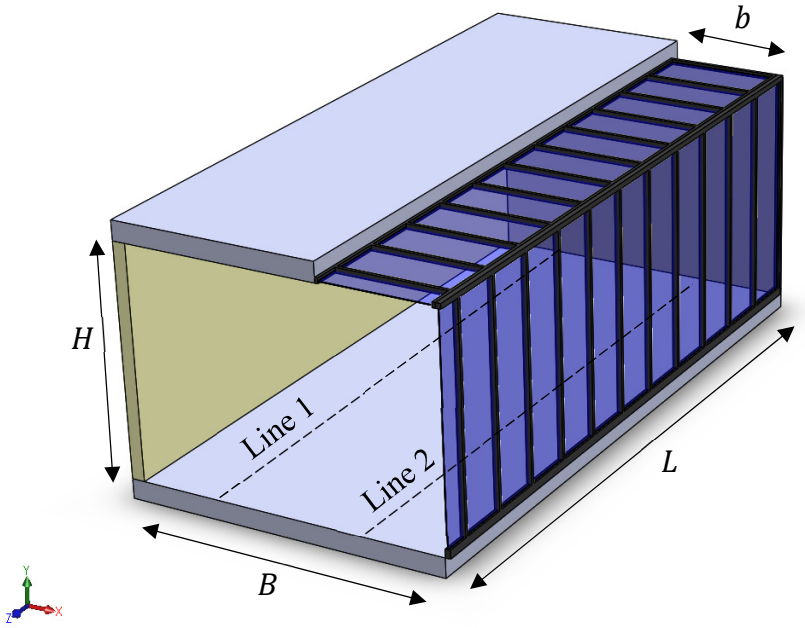


Figure 1 Room dimensions (not drawn to scale).

Table 1 Semi steady-state surface temperatures at 16:00.

	Surface	
1	Front end wall (B x H)	27°C
2	Rear long wall (L x H)	28°C
3	Rear end wall (B x H)	27°C
4	Front long glazed wall (L x H)	45°C
5	Floor (L x B)	22°C
6	Unglazed ceiling (L x $(B - b)$)	28°C
7	Glazed ceiling (L x b)	48°C
8	Chilled ceiling beam (later in the assignment)	

Calculations:

1. Create a self-developed program/script to calculate the solid angle subtended by a rectangular surface **from first principles**. (Perform numerical integration using any programming language or spread sheet of your choice). Attach a print-out of your program script to your submission.

Verify that your program is approximately correct by comparing it with the angle factor figure shown in the lecture slides. Also check whether the angle factors of a full 3D space adds up to 1 **(do not submit your verification)**.

2. Consider a person on "line 1" midway between the two end walls. Assume the person to be positioned at a single point 0.95 m above the floor. Use your script of question 1 to calculate the **mean radiant temperature** [°C], **operative temperature** [°C], **vertical radiant asymmetry** [°C] and **horizontal radiant asymmetry** [°C], for the person. (Tabulate intermediate calculation results by using the above surface names and reference your

answers accordingly). Is this person comfortable? If not, comment on what should be done.

3. Repeat question 2 for a person on "line 2" midway between the two end walls. Compare and discuss the results of question 2 and 3.

In order to reduce the vertical radiant asymmetry on line 2, you are considering the addition of a chilled beam, running along the ceiling next to the glazed section spanning the entire length of the room (see Figure 2). It has a width of b_c and carries chilled water within channels at a uniform temperature of t_c. Ignore temperature gradients in the water or in the beam itself and assume the beam surface to also be at t_c.

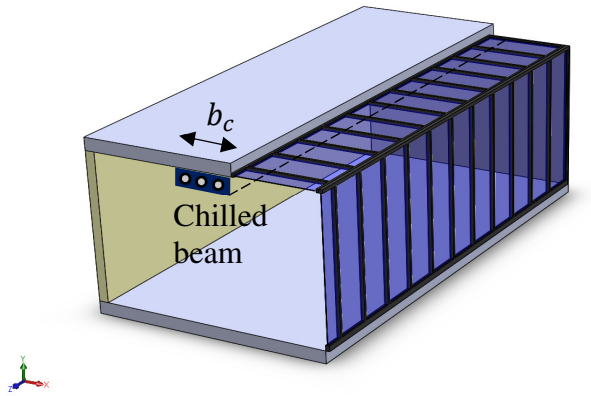


Figure 2 Inclusion of a chilled beam.

4. What is the coldest water temperature that should be considered for this application? Motivate your answer. Show relevant calculations.

For a water temperature of t_c = 11°C (not the correct answer of question 4) do the following:

5. Ignore the height of the chilled beam (in the y direction) by assuming its effect to be the same as a surface flush with the ceiling (surface 8 in Table 1). Determine the influence of b_c on the vertical asymmetry for a person midway along line 2. Plot this dependence. Show data-points, axes markers and axes values. Only show your tabulated angle factor inputs and results for b_c = 1 m (add as many other b_c cases as you want, but do not show their tabulations). Comment on and discuss your results.
6. How has the inclusion of the chilled beam affected the horizonal asymmetry? Supply a relevant plot to support your answer, and offer a discussion.

REMEMBER TO INDICATE HOW MANY HOURS YOU SPENT ON THIS ASSIGNMENT.